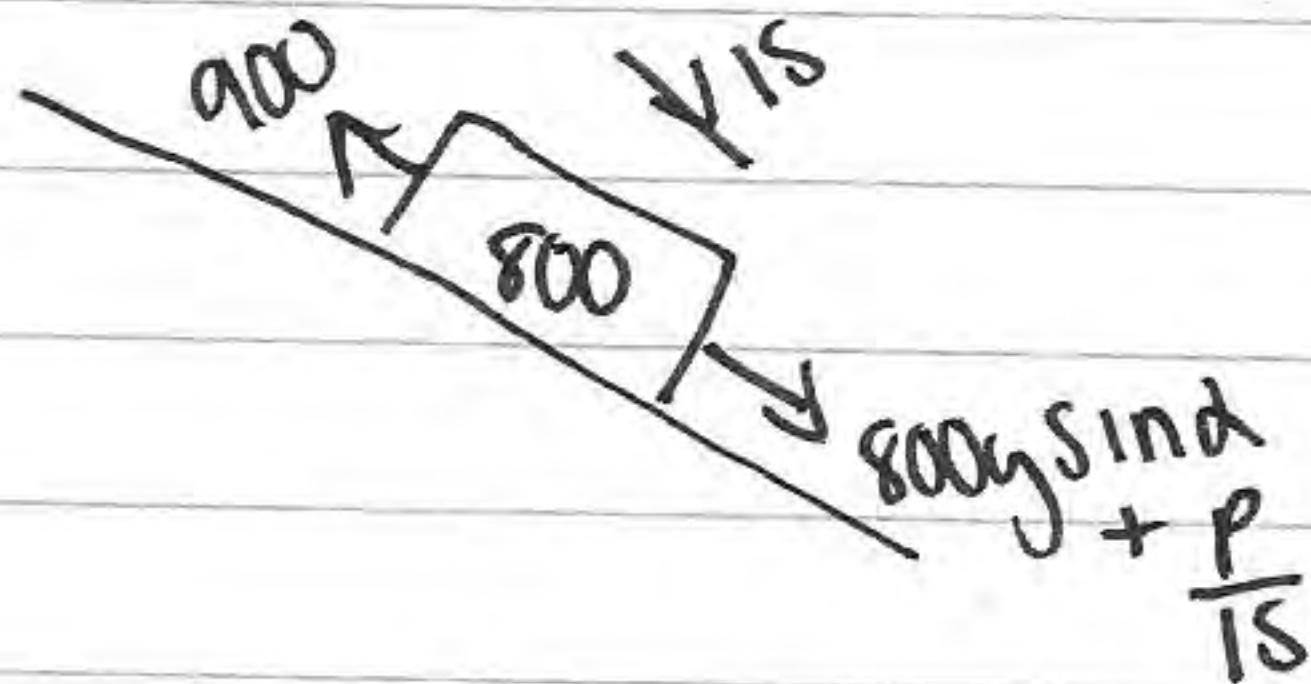


M2 JAN 07

- 1) Wd against friction = $KE_{\text{lost}} = \frac{1}{2}m(v^2 - u^2) = \frac{1}{2}(0.8)(15^2 - 10^2) = 5$
- b) $f_{\text{max}} = \mu NR = \mu 0.8g \Rightarrow \mu 0.8g \times 20 = 50 \Rightarrow \mu = \frac{25}{8g} = \frac{0.31}{(2.5)}$

2)



$$\frac{800g}{24} + \frac{P}{15} = 900$$

$$\Rightarrow \frac{P}{15} = \frac{1720}{3} \Rightarrow P = 8600 \text{ W}$$

$$\Rightarrow \underline{P = 8.6 \text{ kW}}$$

$$Rf \downarrow = ma \Rightarrow \frac{800g}{24} - 900 = 800a \Rightarrow a = -\frac{43}{60}$$

$$u = 15, a = -\frac{43}{60}, v = 0 \quad 0 = 15 - \frac{43}{60}t \Rightarrow \underline{t = 20.9 \text{ sec (3sf)}}$$

3)

$M = 64\pi k$
 $g(16, 0)$

$M = 512\pi k$
 $g(\bar{x}, 0)$

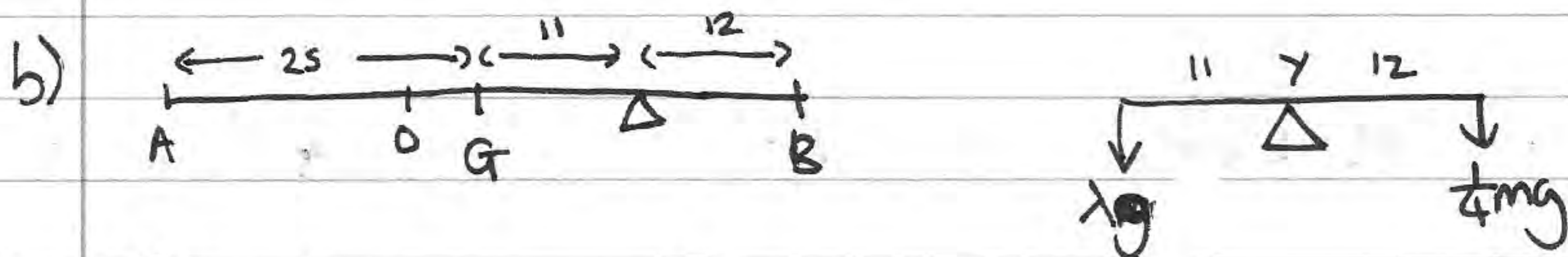
$M = 576\pi k$
 $G(24, 0)$

$k = \text{mass per cm}$

$\uparrow y$
 $\downarrow$

$$64\pi k g \times 16 + 512\pi k g \times \bar{x} = 576\pi k g \times 24 \Rightarrow \bar{x} = \frac{12800}{512}$$

$\bar{x} = 25 \text{ cm}$



$\curvearrowright$

$$\frac{1}{4}mg \times 12 = \lambda g \times 11 \Rightarrow 3m = 11\lambda \Rightarrow \lambda = \frac{3}{11}m$$

4)

before: $P(m) \xrightarrow{u}$, $Q(4m) \rightarrow 0$
 after: $P(m) \xleftarrow{v}$, $Q(4m) \xrightarrow{3v}$

$$e = \frac{4v}{u} = \frac{1}{2} \Rightarrow \underline{u = 8v}$$

b) CLM $\Rightarrow mu = -mV + 4m3v$
 $\Rightarrow 8mv = -mV + 12mv$
 $\Rightarrow 9mv = 12mv \Rightarrow \underline{u = 3}$

5) c)

before: $P(m) \xleftarrow{v}$, $Q(3m) \xrightarrow{3v}$, $R(11m) \rightarrow 0$
 after: $P(m) \xleftarrow{v}$, $Q(3m) \xleftarrow{V_R}$, $R(11m) \rightarrow V_R$

$$e = \frac{2V_R}{3v}$$

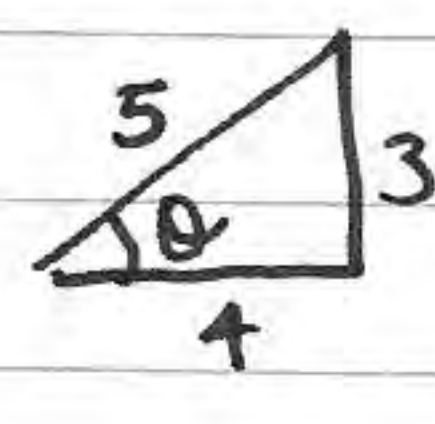
CLM_{QR} $\Rightarrow 9mv = -3mV_R + 11mV_R$
 $\Rightarrow 9mv = 8mV_R$
 $\Rightarrow V_R = \frac{9}{8}v$

$\Rightarrow e = \frac{2(\frac{9}{8}v)}{3v} = \frac{18}{24}$

d) Since $V_R > v$ $\frac{9}{8}v > v$ $\Rightarrow e = \frac{3}{4} \neq$
 Q will collide into P .

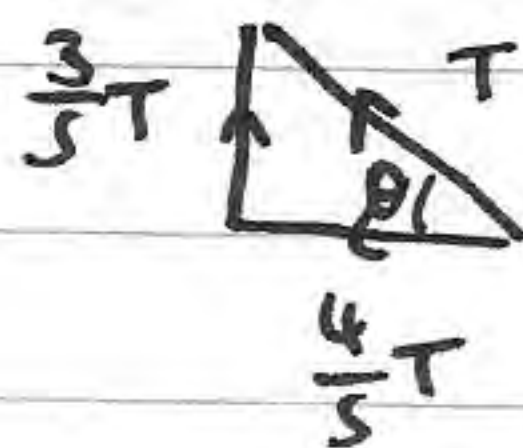
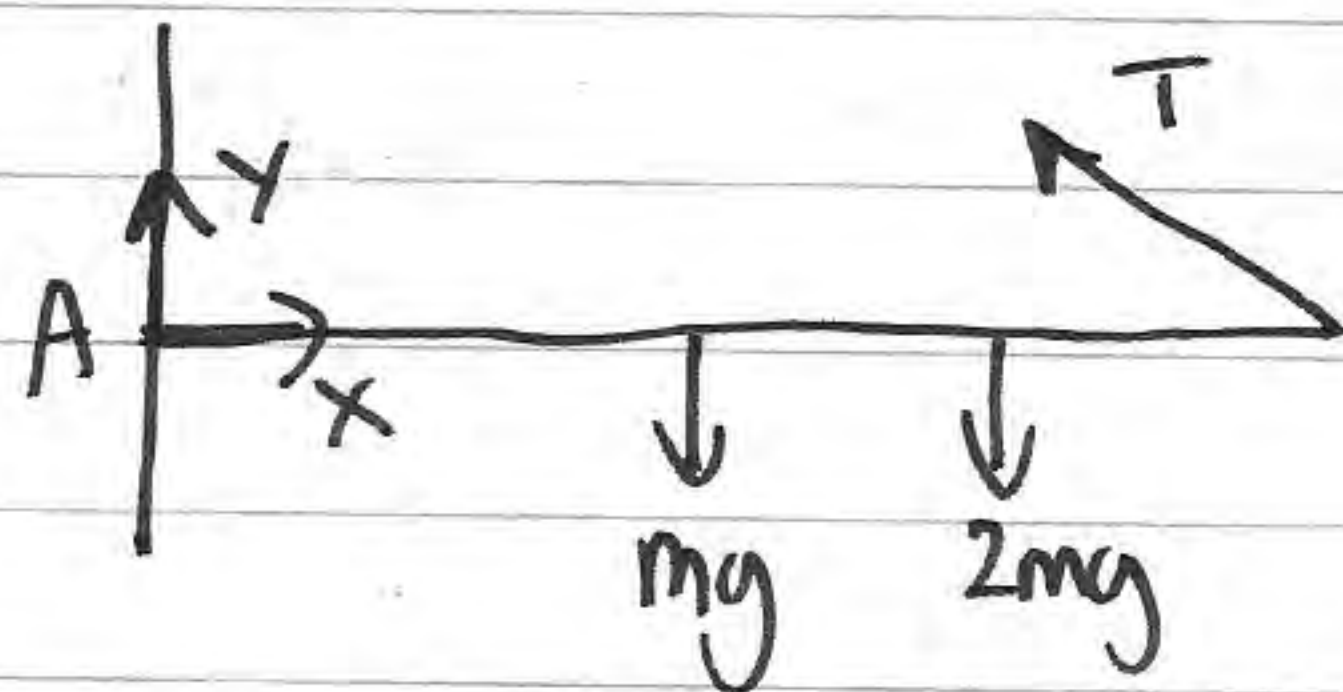
5)

$$\tan \theta = \frac{3}{4}$$



$$\sin \theta = \frac{3}{5}$$

$$\cos \theta = \frac{4}{5}$$



$$\sum F_{\uparrow} = 0 \quad \frac{3}{5}T + Y = 3mg$$

$$\sum F_{\rightarrow} = 0 \Rightarrow X = \frac{4}{5}T$$

$$\sum \tau_A = 0 \quad mg \times 2 + 2mg \times 3 = \frac{3}{5}T \times 4 \Rightarrow 8mg = \frac{12}{5}T$$

$$\Rightarrow T = \frac{40}{12}mg \Rightarrow T = \frac{10}{3}mg \text{ N}$$

$$b) \quad X = \frac{4}{5}T = \frac{4}{5} \left(\frac{10}{3}mg \right) = \frac{40}{15}mg = \frac{8}{3}mg \text{ N} \quad \#$$

$$c) \quad Y = f_{\max} = \mu X = \mu \frac{8}{3}mg$$

$$Y = 3mg - \frac{3}{5} \left(\frac{10}{3}mg \right) = mg$$

$$1mg = \mu \frac{8}{3}mg \Rightarrow \mu = \frac{3}{8}$$

$$6) \quad F = (1.5t^2 - 3)\mathbf{i} + (2t)\mathbf{j} \quad F = ma \Rightarrow a = \underline{(3t^2 - 6)\mathbf{i} + (4t)\mathbf{j}}$$

$$b) \quad v = \int a dt = (t^3 - 6t + c_1)\mathbf{i} + (2t^2 + c_2)\mathbf{j}$$

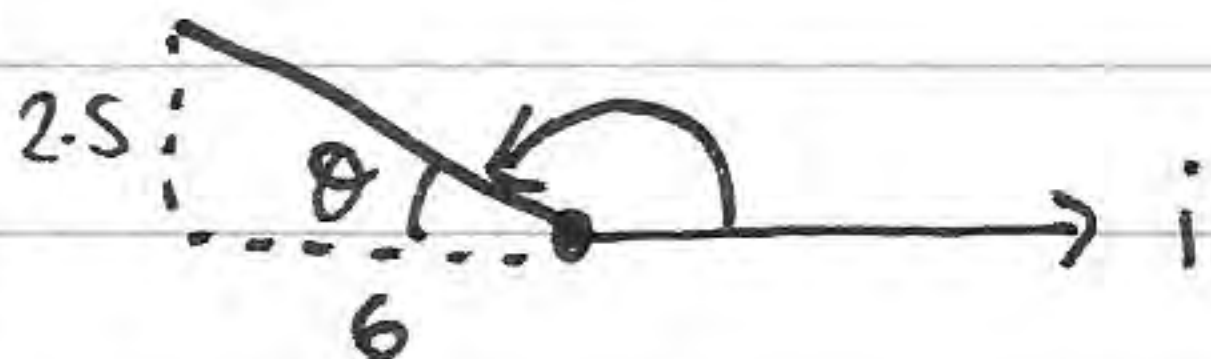
$$t=2 \quad -4i+5j = (-4+C_1)i + (8+C_2)j \Rightarrow C_1=0; C_2=-3$$

$$V = (t^3-6t)i + (2t^2-3)j; \text{ when } t=3 \quad V = (27-18)i + (18-3)j \\ V = 9i + 15j \quad \#$$

c) Momentum before = $4.5i + 7.5j \Rightarrow$ Impulse = $-6i + 2.5j$
Momentum after = $-1.5i + 10j$

$$|Q| = \sqrt{6^2 + 2.5^2} = \underline{6.5 \text{ N s}}$$

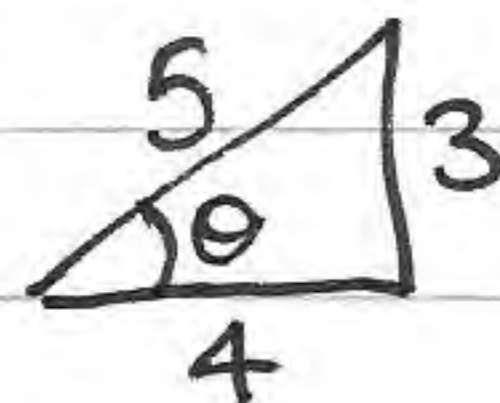
d) $= 180 - \tan^{-1}\left(\frac{2.5}{6}\right) = \underline{157.4^\circ}$



7) Gain in KE = loss in PE
 $\frac{1}{2}m(24.5^2 - u^2) = mg \times 15$

$$\cos \theta = \frac{4}{5}$$

$$\Rightarrow \sin \theta = \frac{3}{5}$$



$$\Rightarrow u^2 = 24.5^2 - 30g$$

$$\Rightarrow u = \underline{17.5 \text{ ms}^{-1}} \quad \#$$

b) $\vec{u} = 17.5 \cos \theta = 14$



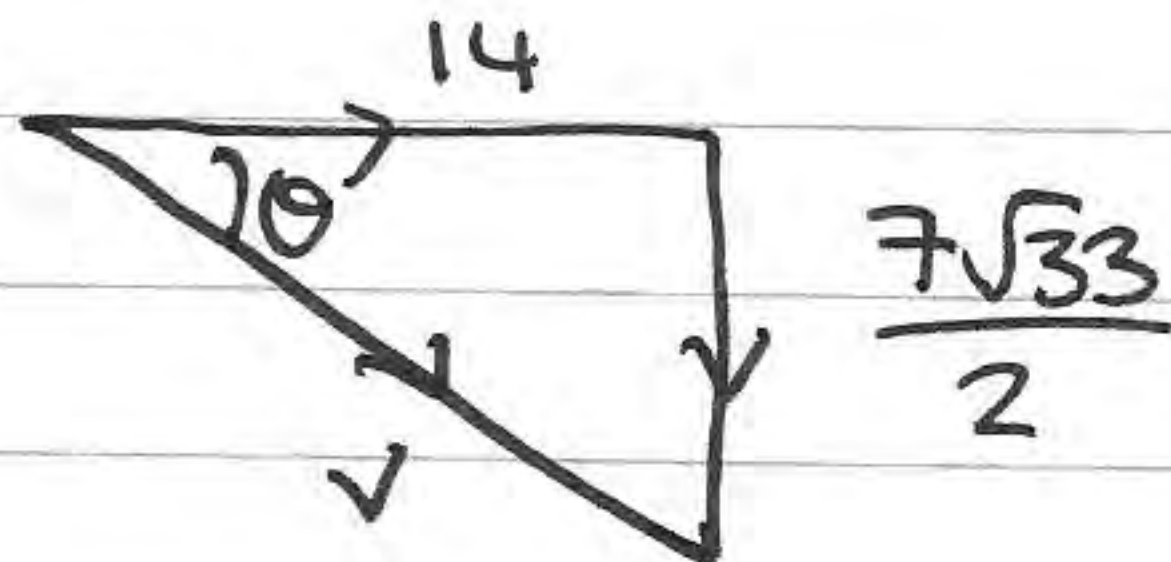
$$u = 17.5 \sin \theta = 10.5$$

$$a = -9.8$$

$$s = -15$$

$$v^2 = 10.5^2 + -19.6x - 15$$

$$v \downarrow = \frac{7\sqrt{33}}{2}$$



$$\theta = \tan^{-1} \left(\frac{\frac{7\sqrt{33}}{2}}{14} \right)$$

$$\theta = 55.2^\circ \text{ (2sf) below H.}$$

c) $u = 10.5$
 $a = -9.8$
 $s = -45$

$$-45 = 10.5t - 4.9t^2 \Rightarrow 4.9t^2 - 10.5t - 45 = 0$$

$$\Rightarrow t = \frac{30}{7}$$



$$u = 14 \quad t = \frac{30}{7}$$

$$BD = 14 \times \frac{30}{7} = \underline{60m}$$